PREDICTORS OF PRONUNCIATION ACCURACY: A REEXAMINATION

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Suter (1976) studied the correlations between English pronunciation accuracy scores and a battery of 20 variables for 61 nonnative speakers of English. Although he found that 12 of the 20 were significantly correlated with pronunciation accuracy, he did not attempt to assess the relative importance of the 12 in accounting for the criterion's variance. By limiting his analysis to a consideration of zero order correlations it was not possible to determine which of the significant predictors were true predictors of pronunciation accuracy in and of themselves and which were only significant predictors by virtue of their substantial correlations with other, more efficient predictors of pronunciation accuracy. In the present paper we apply more advanced techniques to the analysis of the predictors and conclude that only 4 variables, one of them a composite of 2 of Suter's predictors, are useful in accounting for the variability of subjects' pronunciation accuracy scores.

In his study of predictors of pronunciation accuracy, Suter (1976) found that several variables were statistically significant predictors of English pronunciation accuracy. Although he did regress his dependent variable on one predictor variable, first language, his basic approach was that of evaluating zero order correlations between each of the predictors and the dependent variable. But such an analysis does not deal with the problem of correlations between the predictors themselves. If two predictor variables are both correlated with the dependent variable, and if the predictors are correlated with each other, then it may be the case that either of the predictors can successfully predict the dependent variable only insofar as it is correlated with the other predictor. Stated another way, one predictor might be a true predictor of the dependent variable while the other predictor might not really be much of a predictor of the dependent variable in and of itself; it merely appears to be because of its correlation with the true predictor of the dependent variable. All studies which go no further than analyzing zero order correlations suffer from this dilemma. The purpose of the present paper is to reanalyze Suter's data to better determine which of the many predictor variables he examined are the substantive predictors of pronunciation accuracy and which of his predictors do not really contribute to the explanation of pronunciation accuracy. Since Suter's study (1976) comprised the most tightly controlled and comprehensive analysis to date of such a wide range of variables felt to influence learner's pronunciation accuracy, we felt the data worthy of reanalysis with more powerful multivariate techniques. Since these techniques are not widely known in the field of adult second language research, the reanalysis of Suter's data might serve as a good example of the usefulness of such more advanced techniques in the study of adult second language learning.

Analysis

Suter (1976) presented the details of the data acquisition process, which will not be repeated here. We will, however, briefly recap the list of variables he considered. Suter's dependent variable, or criterion, consisted of the average rating by fourteen judges of sixty-one nonnative speakers of English. The judges rated the subjects on a subjective scale of English pronunciation accuracy. Judges' ratings of the subjects were highly correlated. The predictor variables included:

1. Age at which a subject first resided in an English speaking country (AFRE)
2. Age at which a subject was first able to converse meaningfully in English (AFMC)
3. Number of years a subject has lived in English speaking countries (YESC)
4. Percent of conversation at home carried on in English with native speakers of English (%CHE)
5. Percent of conversation at work or school carried on in English with native speakers of English (%CWS)
6. Number of months of residency with native speakers of English (MRNS)
7. Number of years of formal classroom training in English (YFCT)
8. Number of months of intensive formal classroom training in English (MICT)
9. Number of weeks of formal classroom training focused specifically on English pronunciation (WFTP)
10. Proportion of a subject's teachers who were themselves native speakers of English (PTNS)
11. The subject's native language (L1)
12. The number of languages the subject can converse in (NLng)
13. The sex of the subject (Sex)
14. The subject's economic motivation (EcMo)
15. The subject's social prestige motivation (SPMo)

16. The subject's integrative orientation (InOr)
17. The subject's cultural allegiance (CltA)
18. The subject's strength of concern for pronunciation accuracy (SCPA)
19. The subject's aptitude for oral mimicry (AOM)[1]
20. The subject's extroversion or introversion (ExIn)

Each of the twenty predictors had at some time been proposed as a predictor of pronunciation accuracy by others. Except for first language, most of the predictors are either clearly continuous variables (e.g., years, months, weeks) or a psychological measure of some presumed continuous variable (e.g., strength of concern, aptitude for oral mimicry, integrative orientation). One predictor, sex, is dichotomous. First language, as previously noted, was not continuous, but categorical, with four levels.

In his 1976 paper Suter regressed the criterion on the first language predictor (really three dummy coded variables representing the four levels of the categorical variable [see Kerlinger and Pedhazur 1973]) and found that the multiple R for the regression was .65. Since the square of the multiple R, R^2, indicates the proportion of the criterion's variance which is accounted for by the predictors, nearly 42% of the variability of his subjects' pronunciation accuracy could be accounted for by their differences in first language. The next logical step would be to determine how much of the dependent variable's variability could be accounted for by the other predictors, which of them are useful and which of them are not. Ordinarily one could resort to stepwise techniques in the analysis of the usefulness of a battery of predictors in accounting for a criterion's variance, but, as Schmitt et al. (1977) have pointed out, there are dangers involved in using a large number of predictors with a small number of observations of the dependent variable. They note that the ratio of predictors to observations should never be more than 1/5; therefore one should not immediately proceed to regress the sixty-one observations of the criterion on the twenty predictors because the p/o ratio of 20/61 is .33 which is higher than the recommended .2. One could also take the point of view that the twenty predictors do not necessarily represent twenty different variables but possibly represent different measures of a smaller number of underlying variables. This latter point of view could employ factor analysis to derive a smaller number of composite variables

[1] In his 1976 study, Suter collected data on subjects' aptitude for oral mimicry and then adjusted the scores for all subjects to remove differences between languages. It is this adjusted measure that is used in the present study.

from the larger battery of predictors. This smaller number of composite variables could then be used in the stepwise regression analysis of the subjects' pronunciation accuracy scores.[2] This is the approach to be employed in the present analysis.

Table 1 presents a matrix of correlations for the dependent variable and the predictors. No correlation coefficient for the categorical variable first language has been included. The correlations between sex and the other variables are, of course, point biserial correlations, but it makes little difference since the point biserial correlation is merely a Pearson correlation coefficient calculated with dichotomous and continuous variables (Gorsuch 1974, Welkowitz et al. 1976). Note that there are substantial correlations between the predictors themselves. For example, months of intensive classroom training is correlated .75 with weeks of formal training in pronunciation, weeks of formal training in pronunciation is correlated -.41 with age of first meaningful conversation in English, the latter is correlated -.30 with extroversion/introversion, and so on.

The predictor variables (excluding first language) were subjected to a factor analysis using the SAS (version 76.6D) procedure FACTOR, option PRINIT (Barr et al. 1976). PRINIT performs a principal components analysis of the data, extracts as many components as there are variables, and determines the number of components/factors depending on various user specified parameters. We used the default thresholds of minimum eigenvalues of 1.0 and maximum number of factors equal to the number of variables in the analysis. The initial principal components analysis with ones on the diagonal of the correlation matrix chose 8 components/factors. Next, PRINIT made 20 iterations with varying estimates of the communalities of the variables on the diagonal of the correlation matrix until it was satisfied that the eight factors adequately represented the correlations between the nineteen variables. This analysis assumes that we are interested in discovering which of the nineteen variables might be combined into new composite variables based on their correlations, and therefore aims at accounting only for the correlations between variables, not for the entire variance of the variables, as does principal components analysis.

Next, the eight factors extracted by PRINIT were subjected to orthogonal rotation with method Varimax, which seeks to derive simple struc-

[2] Both Kerlinger and Pedhazur (1973) and Gorsuch (1974) have good discussions of this approach.

Table 1
Intercorrelation matrix of Pearson correlation coefficients for dependent variable pronunciation accuracy and 19 predictor variables. N = 61.

	PrAc	AFRE	AFMC	YESC	%CHE	%CWS	MRNS	YFCT	MICT	WFTP	NLng	PTNS	Sex	EcMo	SPMo	InOr	CltA	SCPA	AOM
AFRE	-.227																		
AFMC	-.356	.455																	
YESC	.322	-.288	-.031																
%CHE	.267	-.241	-.041	.026															
%CWS	.396	-.237	-.213	.096	.311														
MRNS	.217	-.338	-.220	.582	.179	.034													
YFCT	-.233	.053	-.185	-.005	-.246	-.253	.274												
MICT	.262	-.171	-.214	.185	.173	.157	.353	.077											
WFTP	.154	-.259	-.412	.091	.082	.181	.388	.387	.750										
NLng	.206	-.003	-.291	-.155	.288	.322	-.033	-.084	-.083	-.006									
PTNS	.189	-.097	.049	.229	.001	.231	.075	-.199	.075	.120	.121								
Sex	-.097	.072	.017	-.082	-.041	-.123	.077	-.040	.122	-.005	.044	.088							
EcMo	.071	-.006	.002	.154	.006	.023	.151	-.007	.244	.037	-.034	.021	.125						
SPMo	.174	-.052	.027	.174	-.148	-.050	.217	.109	.103	.050	-.168	-.050	-.056	.382					
InOr	-.270	.215	.275	-.302	.132	-.032	-.409	-.102	-.159	-.233	-.085	-.141	-.125	-.263	-.450				
CltA	.044	-.001	.188	-.028	.035	.019	.034	.185	.009	.113	.053	.025	-.225	.005	.095	.255			
SCPA	.470	-.181	-.156	.234	.151	.218	.268	.055	.148	.182	.179	-.049	.143	-.031	.033	-.263	.025		
AOM	.330	.063	-.173	.178	.067	-.094	.100	.035	.223	.095	.126	-.114	.181	.247	.144	-.022	.040	.298	
ExIn	.201	-.164	-.300	.035	.188	.062	.107	.024	.046	.095	.084	.053	.045	.050	-.104	-.052	-.017	.049	.181

Variable First Language has been omitted because it is categorical.

Key to abbreviations: PrAc = pronunciation accuracy, AFRE = age of first residence in an English speaking country, AFMC = age of first meaningful conversation in English, YESC = years in an English speaking country, %CHE = percent of conversation at home in English, $CWS = percent of conversation at work or at school in English, MRNS = months of residency with a native speaker of English, YFCT = years of formal classroom training in English, MICT = months of intensive classroom training in English, WFTP = weeks of formal training in English pronunciation, NLng = number of languages spoken, PTNS = proportion of teachers who were native speakers of English, EcMo = economic motivation, SPMo = social prestige motivation, InOr = integrative orientation, CltA = cultural allegiance, SCPA = strength of concern for pronunciation accuracy, AOM = aptitude for oral mimicry, ExIn = extroversion/introversion.

ture by dealing with all of the loadings for all of the variables across a single factor at one time.[3] The data were also subjected to Quartimax and Equamax orthogonal rotations, which each derived nearly identical patterns of the same variables loaded on the same factors with nearly the same loadings. The fact that the three orthogonal rotations produced substantially the same results indicates that the data do indeed exhibit simple structure (Gorsuch 1974). The data were also subjected to SAS's Procrustean (oblique) rotation which also derived the same pattern of salient loadings of the same variables on the same factors as did the three orthogonal rotation methods. The correlations between these non-orthogonal factors were slight, with all but one on the order of .1 to .2. Since there did not seem to be any real gain in achieving simple structure using oblique rotation, since the oblique results substantially matched the orthogonal results, and since orthogonal predictors are always more desirable in regression, it was decided to proceed with the results produced by Varimax, which are presented in Table 2.

The loadings for each variable on each factor in Table 2 were evaluated using a procedure presented by Gorsuch. A threshold value equal to twice the Pearson correlation coefficient for the appropriate degrees of freedom (here 59) and alpha = .05 was calculated at .50. Loadings equal to or greater than .5 were deemed salient. Thus factor 1 has 2 salient loadings for the variables Years in an English Speaking Country and Months of Residency with a Native Speaker. Factor 2 has salient loadings for Percent Conversation at Work or School and Number of Languages, etc. Some of the factors seem more easily interpretable: F1 seems to involve residency, F3 seems to involve focused formal training, F4 is Aptitude for Oral Mimicry, F5 involves age at first contact, F6 involves attitude, and F8 is formal training.[4] On the other hand some factors are less easily inter-

[3] Without going into the exact details, simple structure refers to a pattern of loadings (correlations) of variables on factors in a factor pattern matrix. Ideally each variable has a single loading close to 1.0 on one factor, different variables have such large loadings on different factors, and the rest of the loadings of variables on factors are close to 0.0. If such is the case then the matrix is said to exhibit simple structure and its interpretation is relatively simple.

[4] Because of the uncertainties involved in naming factors, some authors suggest that researchers forswear the practice. We will take the less drastic position of naming single variable factors with the name of the single variable which significantly loads on the factor (e.g., factor 8). Factors with more than one variable significantly loaded on them will be given simple names if there seems to be a clear basis for such a name (e.g., "Residency" for factor 1). Factors with less than a clear basis for naming will be left unnamed. The same holds for the composite variables to be derived below.

Table 2
Rotated factor pattern matrix (Varimax rotation) for 19 variables and 8 factors

Predictors	Factors 1	2	3	4	5	6	7	8	h^2
AFRE	-.369	-.116	-.111	.172	.502*	-.035	-.135	.130	.481
AFMC	.020	-.152	-.152	-.054	.911*	-.052	.093	-.166	.920
YESC	.769*	.008	.024	.074	-.038	.158	.027	-.094	.634
%CHE	.113	.359	.132	.084	-.102	-.181	.134	-.253	.291
%CWS	.040	.566*	.135	-.142	-.173	.041	.183	-.250	.486
MRNS	.711*	.072	.244	.020	-.149	.151	-.038	.211	.662
YFCT	.067	-.195	.142	.017	-.071	.016	.117	.832*	.776
MICT	.147	.031	.925*	.127	-.062	.135	-.027	-.081	.924
WFTP	.125	.128	.797*	-.051	-.257	.048	.093	.321	.851
NLng	-.194	.668*	-.107	.145	-.174	-.059	-.093	.058	.563
PTNS	.168	.331	.054	-.118	.017	.022	-.101	-.100	.176
Sex	.019	.060	.021	-.056	.094	-.005	-.518*	.043	.287
EcMo	.071	-.014	.138	.170	.051	.445	-.066	-.085	.265
SPMo	.110	-.102	.000	-.002	.061	.807*	.151	.073	.706
InOr	-.307	-.145	-.040	.075	.217	-.601*	.364	-.169	.693
CltA	.209	.054	.065	.003	.197	-.033	.507*	.166	.333
SCPA	.304	.372	.071	.210	.067	.039	-.088	.114	.307
AOM	.069	.008	.072	.975*	-.120	.164	.138	.002	1.021
ExIn	.068	.079	.037	.151	-.290	-.082	-.030	-.014	.127
Eignv	3.125	1.716	1.333	1.180	1.067	0.817	0.653	0.613	
Cntrb	29.8%	16.4%	12.7%	11.3%	10.2%	7.8%	6.2%	5.9%	
Cumul	29.8%	46.2%	59.0%	70.2%	80.4%	88.2%	94.4%	100.0%	

Salient loadings are indicated with asterisks. Also shown are the final communality estimates (h^2) for each of the variables and the eigenvalues for each of the factors as computed at the completion of the 20 iterations of the principal axes method of factor extraction. Key to abbreviations: PrAc = pronunciation accuracy, AFRE = age of first residence in an English speaking country, AFMC = age of first meaningful conversation in English, YESC = years in an English speaking country, %CHE = percent of conversation at home in English, %CWS = percent of conversation at work or at school in English, MRNS = months of residency with a native speaker of English, YFCT = years of formal classroom training in English, MICT = months of intensive classroom training in English, WFTP = weeks of formal training in English pronunciation, NLng = number of languages spoken, PTNS = proportion of teachers who were native speakers of English, EcMo = economic motivation, SPMo = social prestige motivation, InOr = integrative orientation, CltA = cultural allegiance, SCPA = strength of concern for pronunciation accuracy, AOM = aptitude for oral mimcry, ExIn = extroversion/introversion, Eignv = eigenvalues, Cntrb = percentage of variance accounted for by this factor, Cumul = the cumulative percentage of variance accounted for by this together with the preceding factors.

preted: F2 involves Percent Conversation at Work or School and Number of Languages, while F7 deals with Sex and Cultural Allegiance. It seems pointless to agonize over succinct labels for the eight factors for we are not interested in the factors themselves, but rather in their ability to account for the variance of the dependent variable.

Since we resorted to factor analysis to derive composite variables from the original set and not to extract factors, the new composite variables were calculated in a manner recommended by Gorsuch. All of the variables were first standardized to mean = 0 and standard deviation = 1. Each new composite was calculated as the algebraic sum of those variables which had salient loadings for that factor in Table 2. For example, composite variable 1 equalled the sum of Years in an English Speaking Country and Months of Residency with a Native Speaker, which were in standard score form. Composite 6 equalled Social Prestige Motivation minus Integrative Orientation, again in standard scores. Composite 8 really was just Years of Formal Classroom Training in standard score form. The rationale behind ignoring the nonsalient variables in the calculation of the new composites is that they represent chance correlations between the original variables and the "factors" and to include them in the calculation of the composites would be to capitalize on chance. Gorsuch also notes that studies have shown that to include variables with nonsalient loadings in the calculations of the composites yields results that are further from the true population parameters than the results obtained if the nonsalient variables are excluded. By limiting ourselves to variables with salient loadings, we exclude the chance effects of extraneous variables.

Before taking up the regression analysis of the newly derived composite variables, it is necessary to check one further item. Factor analysis lumps together variables which are mutually correlated. There exists the possibility that a variable could be unrelated to (uncorrelated with) the other variables included by the factor analysis. Variables with very low communality estimates tend to have very little shared variance with the other variables. But just because they have little in common with the other variables does not say anything about their ability to account for pronunciation accuracy. Therefore the five variables with no salient loadings on any of the factors of Table 2 (Percent Conversation at Home in English, Proportion of Teachers who were Native Speakers, Economic Motivation, Strength of Concern for Pronunciation Accuracy, and Adjusted Extroversion/Introversion) were checked for their correlations with the dependent variable. Two of them (Percent Conversation at Home in English and

Strength of Concern for Pronunciation Accuracy) were found to have significant correlations with Pronunciation Accuracy on Table 1 (critical value = .25 for 59 df at alpha = .05). These two variables, therefore, should be included in our regression. Table 3 presents a correlation matrix for the dependent variable, the composite variables, and the two significant variables not included in the composites.

Having determined which variables are to be included, we may proceed with the regression analysis. The dependent variable Pronunciation Accuracy was regressed on First Language, the eight composite variables, Percent Conversation at Home in English, and Strength of Concern for Pronunciation Accuracy using procedure STEPWISE, option MAXR (maximum R^2 improvement) of SAS (Barr et al. 1976). The maximum R^2 improvement option differs from normal stepwise procedures in that it searches all predictors for the best 1-predictor model, then searches all predictors (not all remaining predictors) for the best 2-predictor model, then searches for the best 3-predictor model. Any variable entered on a previous step may be thrown out on a subsequent step unless the analysis is somehow constrained. Since Suter (1976) had shown that First Language alone accounted for 42% of the variance of the subjects' pronunciation accuracy scores, since an inspection of Table 3 showed that no other predictor would match or exceed the contribution of variable First Language, and since First Language was represented by three dummy coded variables, it was decided to force the three First Language predictors into the analysis first and to constrain the analysis to retain all three in all subsequent steps. The analysis was unconstrained with regard to the ten other predictors.

Table 4 presents the results of the stepwise analysis of the data. Part A indicates that the full 11-predictor model is significant at alpha = 0.0001, which indicates that we are free to proceed with the analysis of individual steps. (SAS procedure MAXR quit after the inclusion of eleven predictors; the twelfth variable would not further improve R^2.) Part B presents the results for each of the steps. First Language accounts for nearly 42% of the variance of subjects' pronunciation accuracy scores and is the best 1-predictor model.[5] First Language and composite variable 4, really Aptitude for Oral Mimicry, together comprise the best 2-predictor model, accounting for 55.9% of the criterion's variance. The 2-variable model represents an improvement of 14.1% over the 1-predictor model. The

[5] Although the three dummy coded variables physically are three predictors, they are conceptually one variable, L1, and will be treated as such in our analysis.

Table 3
Matrix of correlations between the dependent variable, the eight composite variables, percent of conversation at home in English, and strength of concern for pronunciation accuracy. N = 61.

	PrAc	CV1	CV2	CV3	CV4	CV5	CV6	CV7	CV8	%CHE
CV1	.303									
CV2	.370	-.250								
CV3	.222	.306	.082							
CV4	.330	.156	.020	.170						
CV5	-.342	-.289	-.268	-.331	-.064					
CV6	.261	.364	-.036	.171	.098	-.178				
CV7	-.034	-.004	-.018	-.056	-.142	-.037	.085			
CV8	-.233	.151	-.207	.248	.035	-.077	.124	-.144		
%CHE	.267	.115	.368	.137	.067	-.165	-.165	-.049	-.246	
SCPA	.470	.282	.244	.176	.298	-.197	.174	.076	.055	.151

Table 4
Results of the stepwise regression analysis of pronunciation accuracy scores

A. Significance of the 11-predictor model:
$F = 8.92,\ P = 0.0001$

B. Squared multiple correlation at various steps:

Step	Predictors	R^2	Increase	F ratio	P
1	First Language	41.761%	41.761%		
2	Aptitude for Oral Mimicry	55.878%	18.545%	18.557	.01
3	Composite Var. 1 (Residency)	63.227%	11.409%	11.391	.01
4	Strength of Concern Pron. Acc.	67.342%	7.047%	7.056	.01
5	Composite Var. 8 (YFCT)	68.780%	2.537%	2.533	ns
6	Composite Var. 2 (%CWS, NLng)	69.722%	1.676%	1.680	ns
7	Composite Var. 5 (AFRE, AFMC)	70.440%	0.718%	1.287	ns
8	Percent Convers. Home English	70.918%	0.478%	0.855	ns
9	Composite Var. 3 (MICT, WFTP)	71.039%	0.121%	0.213	ns
10	Composite Var. 6 (SPMo, InOr)	71.112%	0.073%	0.126	ns
11	Composite Var. 7 (Sex, CltA)	71.157%	0.045%	0.076	ns

C. Statistics for predictors at step 4:

Predictors	beta	F	P
First Language*			
Aptitude for Oral Mimicry	0.287	12.07	.001
Composite Variable 1 (Residency)	0.136	8.17	.006
Strength Concern Pron. Accuracy	0.227	6.80	.012

Estimated population R^2 = 60.001%
Standard error of the estimate = 0.5915
Nonsignificant predictors: F2, F3, F5, F6, F7, F8, and %CHE.

A shows the significance of the all-predictor model, that is, the F ratio for the test H0: $R^2 = 0$. B shows the squared multiple correlations at each step, the increase in the R^2 at each step, the F ratio for the increase in the R^2, and its associated alpha probability. C lists standardized regression coefficients (beta), the F ratio at step 4 for the test of beta = 0, the alpha probability of the F ratio, the estimated population R^2, the standard error of the estimate, and a list of the nonsignificant predictors.

*No standardized regression coefficient (beta) is shown for predictor First Language because the latter really was three dummy coded variables even though we conceptually treated them as one variable. The betas for each of the dummy coded variables have not been listed because their interpretation would be less than straightforward.

best 3-predictor model included First Language, Aptitude for Oral Mimicry, and CV1, a composite variable for residency in an English environment. This 3-predictor model accounted for 63.2% of the criterion's variance, an improvement of 11.4%. The best 4-predictor model included First Language, Aptitude for Oral Mimicry, CV1, and Strength of Concern for Pronunciation Accuracy, which jointly accounted for 67.3% of the criterion's variance. Using a test for the significance of the increase in R^2 resulting from the addition of a predictor to a regression model presented by Kerlinger and Pedhazur, it was determined that the 4-predictor model was the final significant improvement; i.e., the addition of the fifth variable, composite variable 8, to the regression did not significantly improve our ability to account for the criterion's variance. This implies that the 4-predictor model of First Language, Aptitude for Oral Mimicry, CV1, and Strength of Concern for Pronunciation Accuracy does as good a job of accounting for the criterion's variance as does the all-predictor model. Thus the other variables (CV8, CV2, CV5, Percent of Conversation at Home in English, CV3, CV6, and CV7) do not contribute significantly to our ability to account for the criterion's variance. They are useless, once we have taken into account the first four predictors. The implication is that they are correlated with the criterion only as a result of their correlations with the four meaningful predictors of pronunciation accuracy. Part C lists various statistics for the 4-predictor model. The magnitude of the standardized regression weights (betas) gives an additional indication of the relative importance of each predictor in accounting for the criterion's variance. The R^2 adjusted for degrees of freedom (Rozeboom 1978) gives an estimate of the population R^2, i.e., how much our sample R^2 would shrink if these predictors were applied to the population. The estimated population R^2 of 60% indicates that the four meaningful predictors really do account for a substantial portion of the criterion's variance in the population.

The stepwise results indicate that only four of the predictors are useful in accounting for the variance of subjects' pronunciation accuracy scores. First Language was by far the most important predictor accounting for more variance than any other variable. Aptitude for Oral Mimicry was the second most important predictor. The third most important predictor was a composite variable, CV1, the only composite variable to contribute significantly to the regression. CV1 combined Years in an English Speaking Country and Months Residing with a Native Speaker of English. We interpret CV1 as reflecting length of residency in an English speaking environment. The last predictor to contribute significantly to the explanation of the criterion's variance was Strength of Concern for Pronunciation Accuracy. The remainder of the predictors were deemed useless, after the above four were taken into account.

Suter (1976) was not certain exactly what to make of the variable First Language. His data clearly showed that First Language was a good predictor of pronunciation accuracy, but it was not clear what one should make of this finding. Is this finding due to the four languages sampled, or would it hold true for a larger number of languages? Because of the uncertainty connected with the interpretation of the results for First Language, it was decided to conduct an additional analysis in which the effects of First Language were regressed out of the dependent variable. Such an approach treated First Language as a "noise" variable, perhaps not of real interest, one which prevented the inclusion of "theoretically more interesting" variables in the regression. If the noise variable were removed, certain other variables of greater theoretical interest might emerge as useful in a subsequent analysis. After the variance for First Language was regressed out of the criterion, the residuals of the regression were used as a new criterion which was again regressed on the ten composite and predictor variables.[6] The results of this analysis did not markedly differ from the analysis presented in Table 4. The same significant predictors and composite variables entered in the same order. After the inclusion of Aptitude for Oral Mimicry, CV1, and Strength of Concern for Pronunciation Accuracy, the remaining variables were useless in further accounting for the criterion's variance. Thus it really appears that the other variables, be they composites or simple, are of no use in accounting for our subjects' pronunciation accuracy scores.

Discussion

Suter (1976) found twelve variables to be significantly correlated with pronunciation accuracy. His Table 3 listed the twelve variables in descending order of Pearson correlation coefficients. They were:

1. First Language (.65)
2. Strength of Concern for Pronunciation Accuracy (.46)
3. Percent of Conversation at Work or School (.40)

[6] Kerlinger and Pedhazur (1973) as well as Harris (1975) have discussions of this analysis of covariance through regression approach.

4. Age of First Meaningful Conversation (-.35)
5. Years in an English Speaking Country (.32)
6. Aptitude for Oral Mimicry (.29)
7. Intensive Classroom Training in English (.26)
8. Formal Classroom Training in English (-.24)
9. Integrative Orientation (-.24)
10. Age of First Residence in English Speaking Country (-.23)
11. Percent of Conversation at Home in English (.23)
12. Residence with Native Speakers of English (.21)

As previously noted, in the present study, only five of the above twelve significantly contributed to the explanation of the variance of subjects' pronunciation accuracy scores. Of these five, two (Years in an English Speaking Country and Months of Residence with a Native Speaker) were combined into a single composite variable by the factor analysis. Note that the four meaningful predictors of pronunciation accuracy are not necessarily the four variables with the highest (in absolute value) correlation coefficients. What happened to the other seven significant variables of Suter's study? Why didn't they come out significant in the present study? The basic answer is that there was nothing left for them to contribute after the first four meaningful predictors of Table 4 entered the regression. After the first four made their contribution, there was nothing left for the other seven to contribute that had not already been dealt with. For example, Percent of Conversation at Work or at School had the third highest correlation in Suter's study, but it is found to be meaningless by the present analysis. The reason is that after First Language and the composite variable of Years in an English Speaking Country and Months of Residency with a Native Speaker of English enter the regression, all of the variance that would have been accounted for by Percent of Conversation at Work or at School has already been taken up by the variables that have already entered the regression. This implies that whatever correlation Percent of Conversation at Work or at School has with Pronunciation Accuracy is really due to its correlation with First Language and composite variable 1 (residency). The same could be said for the other variables which Suter (1976) found to be significant, but which have been found meaningless in the present study: Age of First Meaningful Conversation in English, Months of Intensive Classroom Training in English, Years of Formal Classroom Training in English, Integrative

Orientation, and Percent of Conversation at Home. They all are correlated with Pronunciation Accuracy only as a result of their correlations with the four meaningful predictors: First Language, Aptitude for Oral Mimicry, Residency, and Strength of Concern for Pronunciation Accuracy.

If we view this analysis in the predictive perspective we can attempt a profile of the nonnative speakers who are most likely to pronounce English well. They are native speakers of the "favored" languages (here, Arabic or Persian). They are good oral mimics. They have lived in an English speaking country for a considerable number of years, and for most or all of the time they have resided with a native speaker of English. Finally, they are concerned about the accuracy of their pronunciation of English. If we may be permitted to indulge our imaginations, we might even create a theoretically superior pronouncer. We might call her Nazila (or him Mohammed, since sex is not a meaningful predictor). She is a Persian. She arrived in the United States nine years ago and has remained here ever since. Two years after her arrival, she married Fred, an American, and continues to share a home with him. Although she is a good mimic, Nazila continues to worry about what remains of her "foreign accent." This is all we need to know in order to be fairly certain that Nazila pronounces English very well. Whether or not she has had much or little formal instruction in English or English pronunciation is of no importance.

Conversely, the profile of nonnative speakers who are most likely to pronounce English poorly would involve persons who are native speakers of one of the "nonfavored languages" (here, Japanese or Thai). They are poor mimics. They have recently arrived in an English speaking country and do not reside with any native speakers of English. They are not particularly concerned about their pronunciation accuracy in English.

What are the implications of these findings for those of us involved in the teaching of languages? One of the most obvious relates to the fact that teachers and classrooms seem to have had remarkably little to do with how well our students pronounced English. There were four predictors which measured aspects of formal training (number of years of formal classroom training in English, number of months of intensive formal classroom training in English, number of weeks of formal classroom training focused specifically on English pronunciation, and proportion of the subject's teachers who themselves had been native speakers of English). None of these proved important in accounting for variations in pronunciation accuracy. There are tempering influences, of course (see Suter 1976), but the overall picture cannot be encouraging for the language teacher who feels that the profession has a lot to do with how well students learn to pronounce a second language.

In fact, the variables which turn out to be important seem to be those which teachers have the least influence on. Native language, the most important predictor, results from historical accident. Similarly, aptitude for oral mimicry seems beyond the control of the instructor; it is doubtful that one can make a good mimic out of a naturally poor one. This seems to match the common view that some people simply have a better "ear" for pronouncing languages, and that there is not a great deal language teachers or learners can do about it. Length of residence in a country where the target language is spoken natively is largely beyond the instructor's control. Prolonged residence in a foreign country is typically the result of political, financial, or long term educational or professional motivations, rather than the direct result of wanting to pronounce a second language more accurately. Finally, while strength of concern for pronunciation accuracy might be fortified by an effective teacher, this concern is often the result of personal motivations and attitudes established well before the student enters the classroom.

Further research is needed, especially with the variable First Language, to verify the picture that emerges from the data of the present study. The present study indicates that the attainment of accurate pronunciation in a second language is a matter substantially beyond the control of educators. In Krashen's terms, the important variables determining pronunciation accuracy in English are all acquisition variables; formal learning is almost entirely out of the picture. Because the sixty-one nonnative speakers in the present study were all post critical period learners, the results seem to provide further evidence that "adults, as well as children, are able to acquire language" (Krashen 1979).

A final word might also be said about the lack of usefulness of Integrative, Economic, and Social Prestige Motivation as predictors of Pronunciation Accuracy. Lambert and Gardner and others have made much of such variables in their studies of second language learning. One should note, however, that their studies have been based on correlational and factor analytic approaches without extending their findings into an explanatory framework such as in the present study. Some hold that the results of factor analytic studies should never be an end in and of themselves. They should always be a guide for further analysis. One good way to test the theoretical constructs produced by factor analysis is to use the factors/composite variables as predictors in a multiple regression analysis of some external criterion. The results of the present study indicate that integrative and other such motivational variables contribute negligible information in the explanation of the variance of our subjects' pronunciation accuracy scores. Other variables are much more imporant.

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